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RESEARCH ARTICLE

Centromeric and non-centromeric satellite DNA organisation differs in holocentric *Rhynchospora* species

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Abstract Satellite DNA repeats (or satDNA) are fast-evolving sequences usually associated with condensed heterochromatin. To test whether the chromosomal organisation of centromeric and non-centromeric satDNA differs in species with holocentric chromosomes, we identified and characterised the major satDNA families in the holocentric Cyperaceae species *Rhynchospora ciliata* ($2n = 10$), *R. globosa* ($2n = 50$) and *R. tenuis* ($2n = 2x = 4$ and $2n = 4x = 8$). While conserved centromeric repeats (present in *R. ciliata* and *R. tenuis*) revealed linear signals at both chromatids, non-centromeric, species-specific satDNAs formed distinct clusters along the chromosomes. Colocalisation of both repeat types resulted in a ladder-like hybridisation pattern at mitotic chromosomes. In interphase, the centromeric satDNA was dispersed while non-centromeric satDNA clustered and partly colocalised to chromocentres. Despite the banding-like hybridisation patterns of the clustered satDNA, the identification of chromosome pairs was impaired due to the irregular hybridisation patterns of the homologues in *R. tenuis* and *R. ciliata*. These differences are probably caused by restricted or impaired meiotic recombination as reported for *R. tenuis*, or alternatively by complex chromosome rearrangements or unequal condensation of homologous metaphase chromosomes. Thus, holocentricity influences the chromosomal organisation leading to differences in the distribution patterns and condensation dynamics of centromeric and non-centromeric satDNA.

Keywords Chromocentre · Cyperaceae · Heterochromatin · Holocentric chromosome · *Rhynchospora* · Satellite repeats

Introduction

Constitutive heterochromatin remains highly condensed throughout almost the entire cell cycle. It is gene poor and mainly composed of arrays of highly repetitive DNA sequences, such as satellite DNA (satDNA) (Oboerdoerffer and Sinclair 2007). These repeats are highly abundant and ubiquitous elements of eukaryotic genomes, consisting of arrays of thousands of tandem repetitions of a basic motif. They are usually located in clusters along the chromosomes and colocalise with chromocentres in interphase nuclei (Schmidt and Heslop-Harrison 1998; Guenatri et al. 2004; Hembelen et al. 2007). Often they are associated with the centromeres of most plant and animal species (Melters et al. 2013), as well as to telomeres and interstitial sites (Guarrido-Ramos 2015).

Chromosomes are classified according to their centromere organisation. Monocentric chromosomes, present in most eukaryotes studied so far, have a single size-restricted centromere that appears as a primary constriction and serves as a kinetochore assembly and microtubules attachment site. In contrast, holocentric chromosomes show kinetochore activity

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along almost the entire side surface of the chromosome, visible as a groove along each chromatid in some plant species (Wanner et al. 2015; Marques et al. 2015). This centromere type has evolved independently several times and is present in different plant groups such as Cyperaceae and Juncaceae (Melters et al. 2012; Bureš et al. 2013). Recently, the so-called meta-polycentric chromosomes representing a putative intermediate between these two chromosome types were identified in plants (Neumann et al. 2012, 2016).

Studies concerning the satDNA organisation were mainly done at monocentric chromosomes. In general, short monomers of ~160–180 or ~320–370 bp in length and head-to-tail orientated were identified, although monomers may largely vary in length (Macas et al. 2002; Hembelen et al. 2007; Melters et al. 2013; Weiss-Schneeweiss et al. 2015). SatDNA repeats are preferentially located in subtelomeric and (peri)centromeric heterochromatin and form highly condensed clusters in interphase nuclei (chromocentres) and metaphase chromosomes, such as, e.g. found in mice (*Mus musculus domesticus*), *Urvillea chacoensis* and *Citrus* species (Guenatri et al. 2004; Urdampilleta et al. 2009; Barros e Silva et al. 2010).

In holocentric species, the few reports of satDNA mapping available so far showed a similar distribution. In the plant genus *Luzula* (Juncaceae), dozens of tandem repeats form large chromocentres in interphase and multiple discrete bands in metaphase chromosomes, predominantly at chromosome ends (Haizel et al. 2005; Heckmann et al. 2013). The MBSAT1 repeat from the cabbage moth (*Mamestra brassicae*, Lepidoptera) is clustered on the holocentric heterochromatic Z and W chromosomes (Mandrioli et al. 2003).

However, new findings from beaksedges (*Rhynchospora*, Cyperaceae), challenged the classical view of satDNA distribution. Instead of containing several satDNA repeat families, the genome of *Rhynchospora pubera* is poor in such sequences, but with the predominance of one satDNA family, named Tyba (Marques et al. 2015). In contrast to other satDNAs, Tyba is dispersed in interphase, but with the onset of mitotic condensation, linear satDNA structures along all chromosomes rather than clustered blocks are formed. A detailed CENH3-ChIP analysis (chromatin immunoprecipitation combined with sequencing-ChIP-seq) confirmed Tyba as the first centromeric tandem repeat described from a holocentric species (Marques et al. 2015).

To test whether the chromosomal organisation of centromeric and non-centromeric satDNA differs in holocentric species, we identified and characterised satDNA families in three other holocentric *Rhynchospora* species. While centromeric repeats revealed linear signals at both chromatids, non-centromeric satDNA formed distinct clusters along the chromosomes. Colocalisation of both repeat types resulted in a ladder-like hybridisation pattern at mitotic chromosomes. In interphase, the centromeric satDNAs appeared dispersed while non-centromeric satDNAs clustered and partly colocalised to chromocentres.

Materials and methods

Plant material and genomic DNA isolation

Specimens of *Rhynchospora ciliata* (Vahl) Kükenth ($2n = 10$), *R. globosa* (Kunth) Roem. & Schult. ($2n = 50$) and *R. tenuis* Link ($2n = 2x = 4$ and $2n = 4x = 8$ cytotypes) were collected from natural populations and maintained in the experimental garden of the Laboratory of Plant Cytogenetics and Evolution (UFPE, Recife-PE, Brazil). Genomic DNA was isolated from young leaves using the DNeasy Plant Mini kit (Qiagen).

Next-generation sequencing

Genomic DNA of *R. ciliata*, *R. globosa* and the diploid cytotype of *R. tenuis* was used for next-generation sequencing (NGS). Briefly, the DNA was fragmented by nebulisation and used for the preparation of a paired-end library with fragment length of approximately 500 bp. Reads (100 nt long) were generated by a Illumina HiSeq2000 system at the IPK (Gatersleben, Germany) sequencing facility.

Genome size measurement

Nuclear DNA content measurements of *R. ciliata* were performed using the WPB buffer according to Loureiro et al. (2007) and a Partec CyFlow SL cytometer equipped with the Flomax software (Partec). Three independent DNA estimations of three different individuals were performed at different days. *Glycine max* (L.) Merr. "Polanka" (2C = 2.55 pg) was used as internal standard (Doležel et al. 2007) and the 2C value was calculated as sample peak mean/standard peak mean × 2C DNA content of standard (pg).

Repeat identification

Graph-based clustering of genomic Illumina reads was performed using the Repeat Explorer pipeline on a Galaxy server (Novak et al. 2010; Novak et al. 2013). In total, 1,269,680, 617,000 and 4,766,968 reads were used to identify and characterise the most-abundant repetitive families in the *R. ciliata*, *R. globosa* and *R. tenuis* genomes, respectively. Contigs assembled in clusters representing repeats with abundance >0.01 % of the genome were further examined in silico for a potential tandem organisation by dot-plot (Sonnhammer and Durbin 1996), and the contig sequences were used to design PCR primers. Alternatively, consensus sequences of satellite repeats were generated using k-mer analysis of unassembled reads (Macas et al. 2010) as implemented in TAREAN tool of the RepeatExplorer.

PCR amplification, cloning and sequencing of plasmids

Primer pairs facing outwards were designed for each newly identified repeat family of *R. ciliata* and *R. tenuis* (Online resource 1). They were applied with the primers for Tyba (Marques et al. 2015) to amplify fragments from genomic DNA. PCR reactions were performed in 50 μL samples containing 50 ng of gDNA, 0.1 mM dNTPs, 1× PCR buffer, 0.4 μM of each primer and Platinum *Taq* polymerase (Invitrogen) for 30 cycles of amplification (1 min at 94 °C for denaturation, 1 min at 50 °C for annealing and 1 min at 72 °C for elongation). The amplicons were cleaned using the Nuclei Spin Gel and PCR Clean-up kit (Macherey-Nagel), then ligated into the pJET1.2 vector (Life Technologies) and transformed into SoloPack Gold competent cells (Agilent Technologies). Positive clones were screened by colony PCR and sequenced at the Sequencing Platform of LABCEN/CB/UFPE (Brazil).

Chromosome preparation

Root tips were obtained from plants growing in pots, pretreated with 2 mM 8-hydroxyquinoline at 10 °C for 20 h, fixed in methanol/acetic acid 3:1 (*v*/*v*) and stored at −20 °C. Slides were prepared following a modification of the protocol described in Carvalho and Saraiva (1993). Briefly, meristematic tissue were treated with an enzymatic solution containing 1.4 % cellulase R10, 1.4 % cellulase, 2.0 % pectolyase and 2.0 % cytohelicase in 1× citric buffer for 1–1.5 h at 37 °C and washed in 1× citric buffer. Root tips were placed on a tilted slide (~30°), washed with cold, fresh fixative (methanol/acetic acid, 3:1, *v*/*v*) and fragmented into smaller pieces with the help of needles. Drops of cold fixative were dripped over the macerated tissue, allowing the free cells to spread downwards over the slide. After air drying, the material was incubated in 45 % acetic acid for 5 min.

Fluorescence in situ *hybridisation*

PCR amplified products of *R. ciliata* and *R. tenuis* satellite DNA sequences were labelled with Cy3-dUTP (Jena Bioscience) or Alexa 488-5-dUTP (Life Technologies) using a nick translation mix (Roche). For *R. globosa*, 5′-Cy3 or 5′-FAM-labelled oligonucleotides (Eurofins MWG Operon, http://www.eurofinsdna.com) were used (*Rg*SAT1/1: 5′-GTGTTGTTTCTGCTGAAATAGTGATTTTGAGTAT-3′ and *Rg*SAT1/2: 5′-GTTTATGCTGAAATAGTGATTTTGAGAAGTTTTCA-3′). Additionally, the plasmid p*Ta*71 (Gerlach

Table 1 Characterisation of satellite repeats in selected *Rhynchospora* genomes

Species	Repeat name	Monomer size (bp)	AT content (%)	Genomic abundance (%)	Chromosomal distribution	
					Interphase	Metaphase
R. ciliata	*Tyba*	172	73.3	5.29	Dispersed	Centromeric; linear labelling along each chromatid
	*Rc*SAT2	90	54.4	3.52	Clustered and colocalised to chromocenters	Non-centromeric; clusters along chromosomes; association with DAPI+ bands chromatin
	*Rc*SAT20	79	52.5	0.4	Clustered but not colocalised to chromocenters	Non-centromeric; clusters along chromosomes; not associated with DAPI+ bands
R. globosa	*Rg*SAT1/1	184	65.8	7.91	Clustered and colocalised to chromocenters	Non-centromeric; clusters mainly at chromosome (sub) temini in association with DAPI+ bands
	*Rg*SAT1/2	186	65.6	3.34	Clustered and colocalised to chromocenters	Non-centromeric; clusters mainly at chromosome temini in association with DAPI+ bands
	*Rg*SAT1/35	182	64.8	0.22	N/A	N/A
	*Rg*SAT47	166	45.8	0.12	N/A	N/A
	*Rg*SAT158	209	45.9	0.01	N/A	N/A
	*Rg*SAT173	156	75	0.01	N/A	N/A
R. tenuis	*Tyba*	173	76.3	2.97	Dispersed	Centromeric; linear labelling along each chromatid
	*Rt*SAT239	82	50	0.01	Clustered	Non-centromeric; clusters along larger chromosome pairs or mainly at terminal position on smaller pairs

Except for centromere-specific Tyba repeat, each sequence was found to be species-specific

and Bedbrook 1979), containing the 18S-5.8S-26S ribosomal DNA (rDNA) from wheat, was labelled by nick translation as described above. Pre-treatment of slides and probe denaturation followed Fonsêca et al. (2010). The hybridisation mix contained 50 % (*v*/*v*) formamide, 2× SSC, 5 mM EDTA, 0.2 M Tris-HCl (pH 8.0), 2.8 μg salmon sperm DNA and 200–300 ng of the probe. Slides were denatured for 8 min at 75 °C and hybridised at 37 °C for up to 2 days. Stringent washes were performed in 2× SSC at 58 °C for 20 min. Then the slides were dehydrated in an ethanol series (70, 90 and 100 %) and counterstained with 4′,6-diamidino-2-phenylindole (DAPI) in antifade (Vectashield, Vector). Images were acquired using a Leica DM5500B microscope equipped with a CCD camera (Leica DFC345 FX).

Immuno-FISH

Immuno-fluorescence in situ hybridisation (FISH) was performed as described by Ishii et al. (2015) using *R. pubera*-specific CENH3 antibodies (Marques et al. 2015). The preparations were counterstained as above.

Super-resolution microscopy

To analyse the spatial distribution of satellite repeats beyond the classical Abbe/Raleigh limit at a lateral resolution of ~120 nm (super-resolution, achieved with a 488-nm laser), 3D-structured illumination microscopy (3D-SIM) was applied using a 63×/1.4 Oil Plan-Apochromat objective of an Elyra PS.1 microscope system and the software ZEN (Carl Zeiss GmbH). Images were captured separately for each fluorochrome using the 561-, 488- and 405 nm laser lines for excitation and appropriate emission filters (Weisshart et al. 2016). 3D rendering based on SIM image stacks was done using the Imaris 8.0 (Bitplane) software.

Results

To identify the most-abundant satellite repeats in *R. ciliata*, *R. globosa* and *R. tenuis* an in silico graph-based clustering of genomic Illumina reads was performed according to Novak et al. (2010, 2013). The previously identified centromeric repeat Tyba of *R. pubera* (Marques et al. 2015) was found as the most abundant satDNA among the analysed reads from *R. ciliata* (5.29 % of the genome, 1C = 547.6 Mbp) and *R. tenuis* (2.97 % of the genome; 1C = 381.4 Mbp; Feitoza L, unpublished data), but was not present in *R. globosa*. Additional satellite repeats, varying in abundance but comprising more than 0.01 % of the genomic analysed reads, were identified: two satDNAs for *R. ciliata*, four for *R. globosa* and one for *R. tenuis* (Table 1). *R. globosa* was found to contain one major family, named *Rg*SAT1, which got split into three clusters in our analysis, representing sequence variants

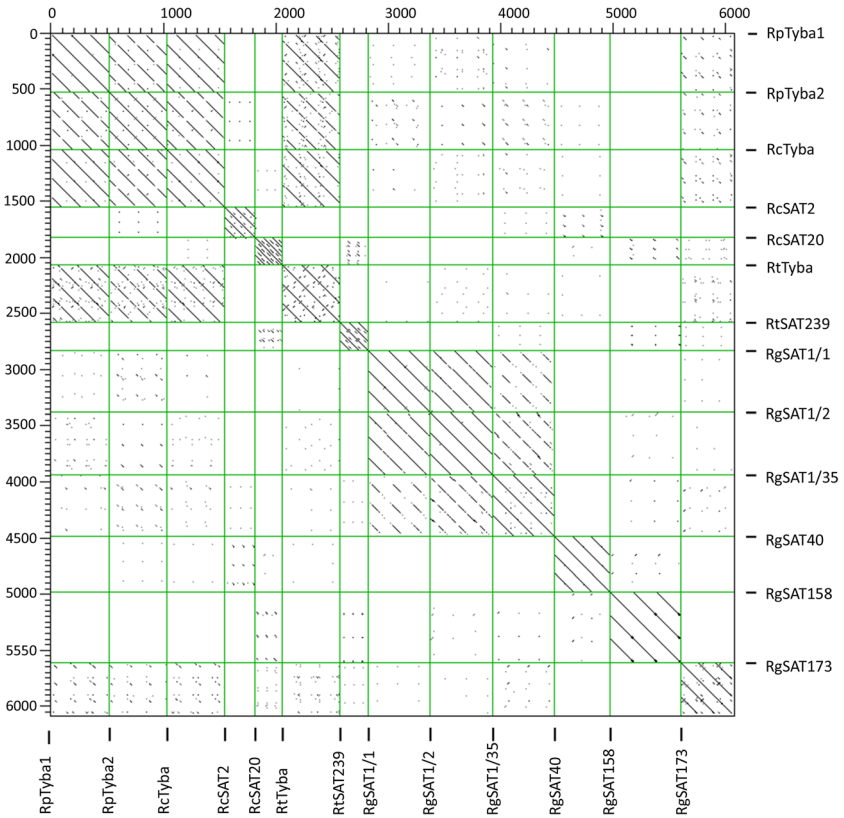


Fig. 1 Dot plot of all satellite repeats identified in three *Rhynchospora* species (*R. ciliata*, *R. globosa* and *R. tenuis*). For the purpose of better visibility on dotter, each sequence was repeated three times and the centromeric repeat Tyba from *R. pubera* (Marques et al. 2015) was also included in the comparison

(subfamilies) of this satellite (Fig. 1). The subfamilies designated as *Rg*SAT1/1 (184 bp monomer), *Rg*SAT1/2 (186 bp monomer) and *Rg*SAT1/35 (182 bp monomer) differed in their genomic abundance (Table 1). *R. ciliata* (*Rc*SAT2, *Rc*SAT20) and *R. tenuis* (*Rt*SAT239) repeats showed complex dot-plot layouts, suggesting the presence of satDNAs with irregular sized repeats/arrays and/or shorter internal subrepeats (Online resource 2). However, monomer sequences could be reconstructed based on frequent k-mer analysis (90, 79 and 82 bp, respectively; Table 1; Online resource 3).

Next, the chromosomal distribution of the in silico-identified repeats was evaluated by FISH. The centromeric Tyba probes (*Rc*Tyba and *Rt*Tyba) resulted in similar distribution patterns in *R. ciliata* and *R. tenuis* as described for *R. pubera* (Marques et al. 2015). In interphase nuclei, the Tyba signals were observed as dispersed foci all over the nuclei (Figs. 2 (a, b) and 3 (a)). With progression of the mitotic cell cycle, the signals were observed as lines along the lateral sides of each chromatid, mirroring the position of the holocentromeres (Figs. 2 (c) and 3 (b)). Figure 2 (c, c') shows the colocalisation between CENH3 and Tyba signals along the entire length of the sister chromatids in *R. tenuis*.

All other satellites showed a clustered distribution in interphase nuclei, but with a different localisation depending on the species. *R. ciliata* and *R. globosa* showed DAPI-enriched ($DAPI^+$, AT-rich) chromocentres (Figs. 3 (a), 4 (a) and 5 (a)), colocalising with *Rc*SAT2 in *R. ciliata* (Figs. 3 (a) and 4 (a)) and *Rg*SAT1/1-*Rg*SAT1/2 in *R. globosa* (Fig. 5 (a, a'); Online movie 1). In *R. globosa*, these two abundant subfamilies showed a complementary labelling in almost all chromocentres, and both *Rg*SAT1/1 and *Rg*SAT1/2 form, in addition to globular organised chromatin fibres, often ring structures in the interphase nuclei (Fig. 5 (a')). In *R. ciliata*, *Rc*SAT20 signals appeared as blocks outside of the chromocentres (Fig. 4 (a)). The *Rt*SAT239 repeat of *R. tenuis* clustered in interphase nuclei and at metaphase chromosomes, but chromocentres could not be identified (Fig. 6).

In contrast to the centromeric Tyba repeats, all satDNAs were mainly organised as blocks located (sub) terminally (*R. globosa*) or distributed along metaphase chromosomes (*R. ciliata* and *R. tenuis*, Table 1). *Rc*SAT2 and *Rc*SAT20 appeared as compact clusters and *Rc*SAT2 colocalised with $DAPI^+$ bands (Fig. 4 (b)). *Rc*SAT20 chromatin was partially decondensed and looped out from the *R. ciliata* metaphase chromosomes (Fig. 4 (b', arrows)). *Rg*SAT1/1 and *Rg*SAT1/2 were detected in almost all chromosome subtermini of *R. globosa* (Fig. 5 (b, b')). In R. tenuis, most of the RtSAT239 signals were enriched along the larger

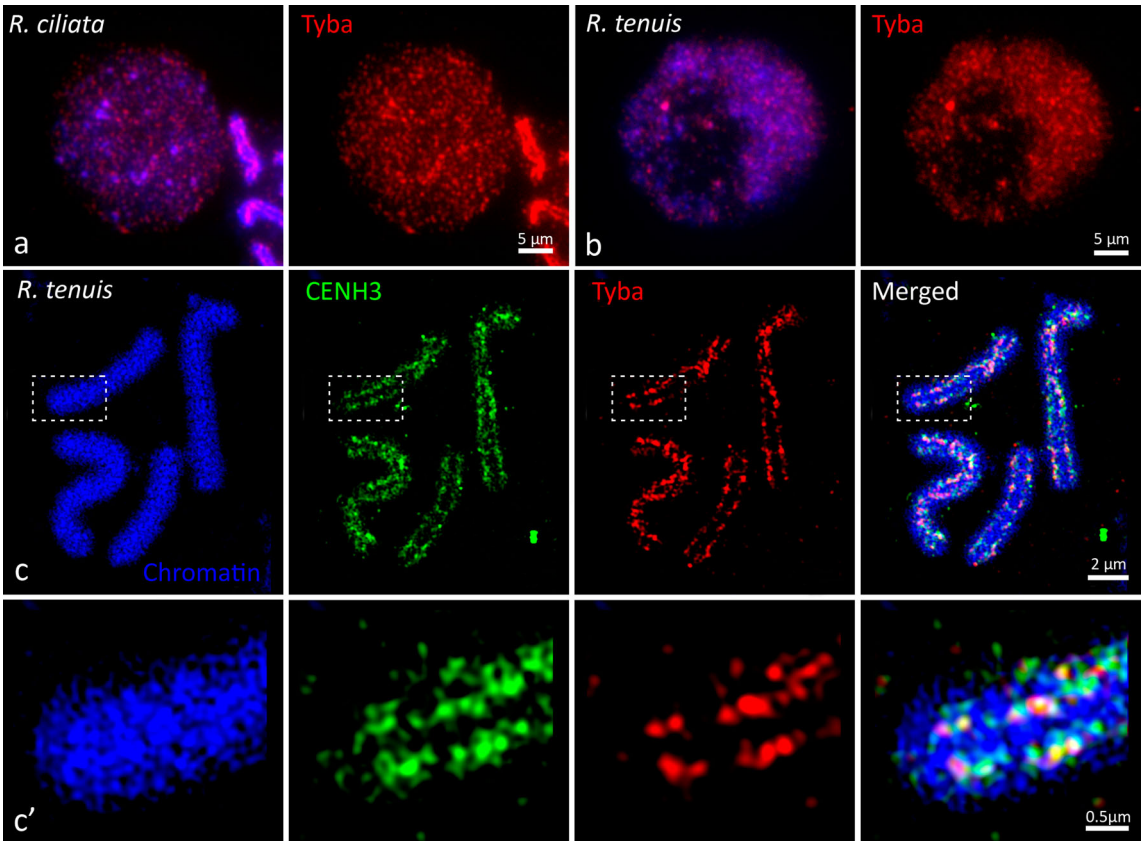


Fig. 2 Distribution of the centromeric satellite DNA Tyba in *R. ciliata* (*a*) and *R. tenuis* (*b*, *c*) analysed by SIM (*c*). Tyba appears dispersed in interphase nuclei (*a*, *b*) but shows a linear distribution and a colocalisation with CENH3 at metaphase chromosomes (*c*, *c'*). *Dotted rectangles* indicate the enlarged region in (*c'*)

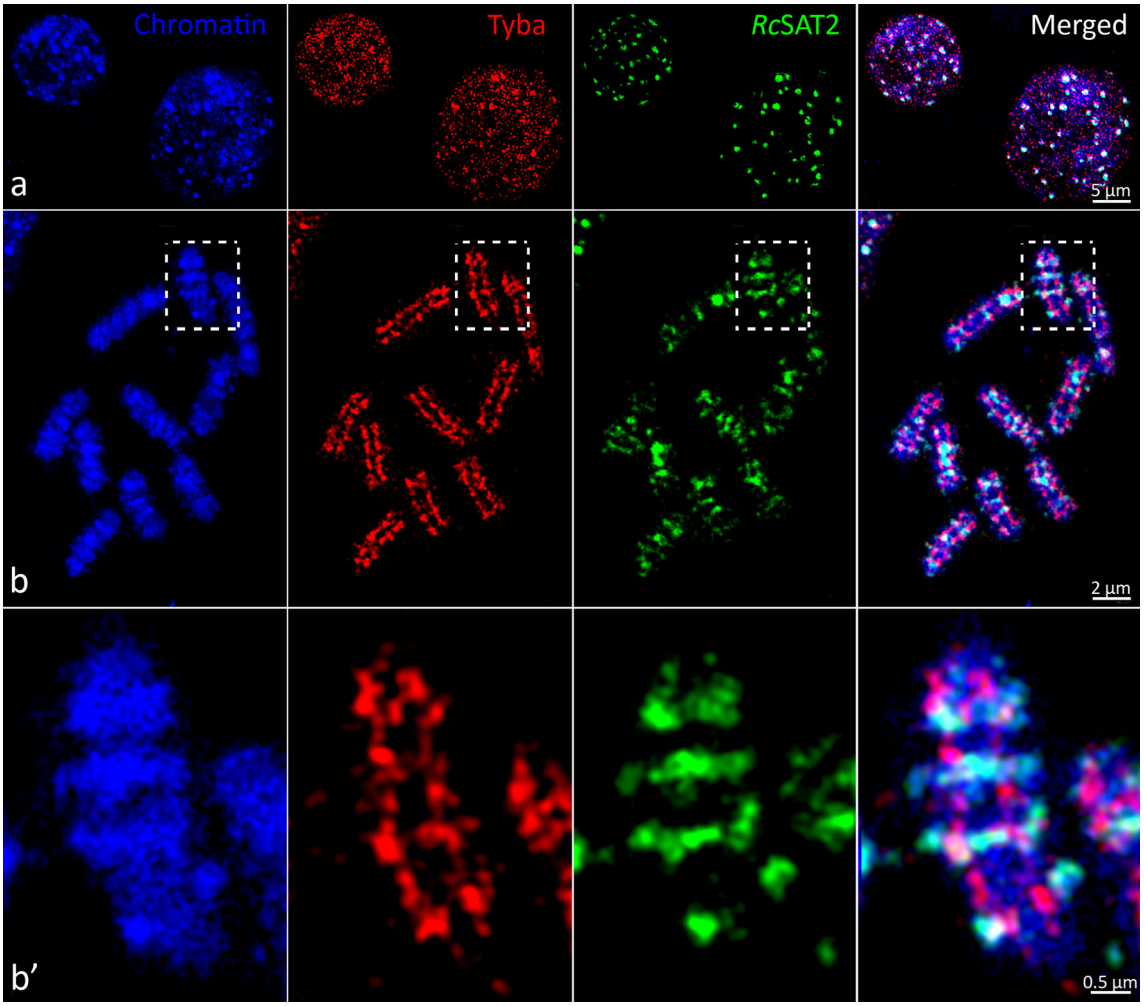


Fig. 3 Comparative distribution of centromeric (*Tyba*) and non-centromeric satellite repeats (*RcSAT2*) in *R. ciliata* analysed by SIM. Tyba is dispersed in interphase nuclei (*a*) but exhibits a linear arrangement, colocalising with the holocentromeres at metaphase chromosomes (*b*). The repeat *Rc*SAT2 forms chromatin blocks at both stages and associates with highly condensed chromatin (*a*, *b*). The combination of both repeats results in a ladder-like hybridisation pattern in mitotic metaphase chromosomes (*c*). *Dotted rectangles* indicate the enlarged region in (*b'*)

chromosomes, while only faint labels were present in the subterminal regions of the smaller chromosomes (Fig. 6 (b)). Similar to *Rc*SAT20, but to a lesser extent, *Rt*SAT239 appeared also decondensed at mitotic metaphase chromatin (Online resource 4, arrowheads). The colocalisation of both centromeric and non-centromeric satDNAs resulted in a ladder-like hybridisation pattern at mitotic chromosomes, as visualised for Tyba and *Rc*SAT2 in *R. ciliata* (Fig. 3 (b, b')).

Despite the banding-like hybridisation patterns of the non-centromeric satDNA, the identification of chromosome pairs was impaired due to the irregular hybridisation pattern of the homologues in *R. tenuis* and *R. ciliata*. In the autotetraploid accession of *R. tenuis*, the expected similar satDNA distribution has not been observed at all large or small chromosomes (Fig. 6 (c); Online resource 4). In *R. ciliata*, even when the 45S rDNA was employed as an additional marker for chromosome identification, the labelling pattern of the homologues was not identical, and the four chromosomes bearing 45S rDNA sites showed a different labelling by the *Rc*SAT2 probe (Online resource 5).

Discussion

The composition of satellite repeats was investigated in three different holocentric *Rhynchospora* species. Besides the holocentromere-localised repeat Tyba present in *R. ciliata* and *R. tenuis*, five species-specific satDNA families were identified in *R. ciliata*, *R. globosa* and *R. tenuis*. The chromosomal localisation showed a contrasting distribution among the different sequences. While centromeric Tyba repeats showed the same dynamic dispersed labelling at interphase and the association with the holocentromeres in metaphase

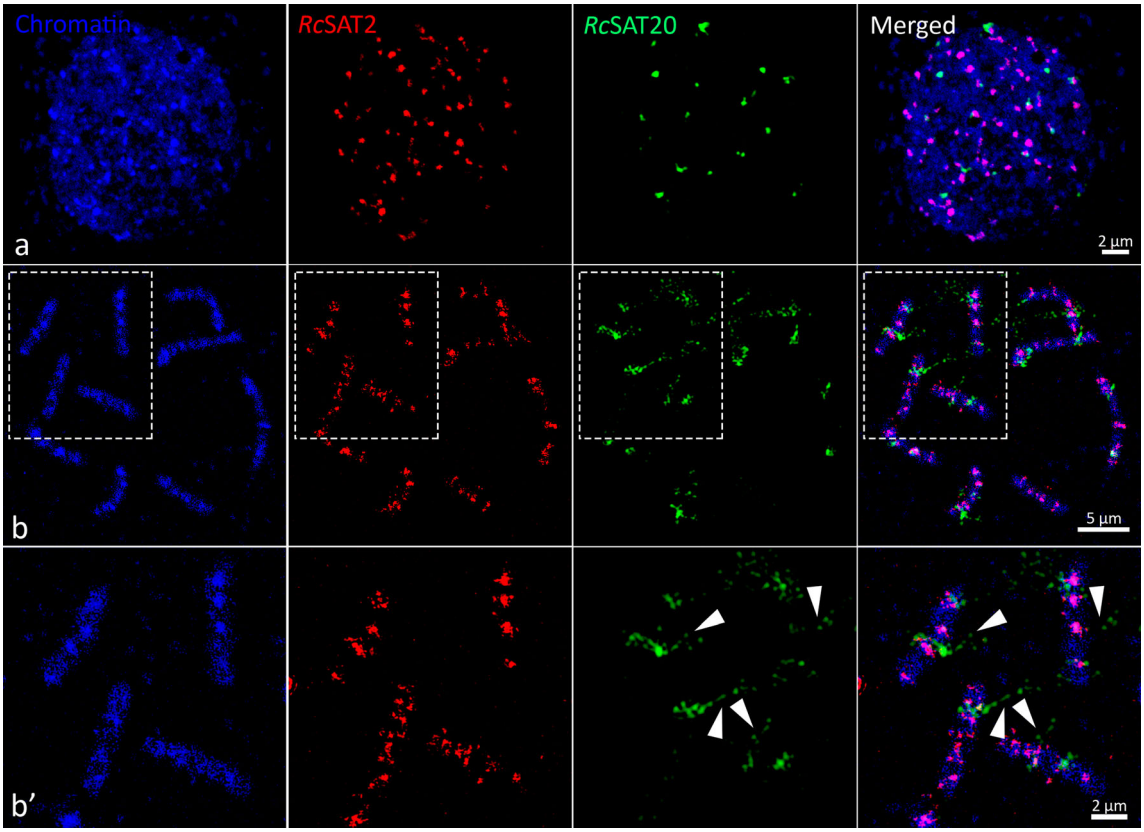


Fig. 4 Comparative localisation of non-centromeric repeats in *R. ciliata* analysed by SIM. The repeats *Rc*SAT2 (*red*) and *Rc*SAT20 (*green*) are both clustered at interphase (*a*) and metaphase (*b*) but occupy distinct domains. Furthermore, *Rc*SAT20 shows partial decondensation in metaphase chromosomes forming chromatin loops (*arrowheads*). The marked area in (*b*) is enlarged in (*b'*)

chromosomes of *R. ciliata* and *R. tenuis*, like in *R. pubera* (Marques et al. 2015), the newly identified non-centromeric satellites exhibited the typical clustered pattern found for satellite DNAs in other monocentric and holocentric species. This distinct arrangement of Tyba compared to other repeats in the same genome reinforces its putative role in centromere function, previously demonstrated by the highest level of association to CENH3 in ChIP-seq analysis in *R. pubera* (Marques et al. 2015). A linear distribution of centromeric repeats along the chromatids might be a feature of species with extended centromeres. Similarly, linear centromeric satDNA clusters are distributed along the extended centromeres of *Pisum* and *Lathyrus* species (Neumann et al. 2012), although in a more restricted chromosomal region.

There are many reports on the distribution of satDNA families in monocentric species (Rosato et al. 2012; Čížková et al. 2013). In all cases, these sequences were found as condensed chromatin regions throughout the mitotic cell cycle. In *R. tenuis*, which shows no chromocentres after DAPI staining, the non-centromeric repeat *Rc*SAT239 was clustered in interphase and located either at the ends of the smaller, or as blocks along the larger chromosome pairs. On the other hand, *R. ciliata* and *R. globosa* showed clear chromocentres colocalising with satellite repeats. Earlier studies demonstrated a contrasting heterochromatin distribution and composition in *Rhynchospora* species (Vanzela and Guerra 2000). While some species like *R. ciliata* exhibited clear fluorescence- or Giemsa C-banding positive blocks along the chromosomes, others like *R. pubera* showed only $CMA^{+}/DAPI^{-}$- and NOR-associated heterochromatin. This suggested great heterochromatin variability among *Rhynchospora* species but in a similar organisation as in monocentric plants (Vanzela and Guerra 2000).

Repetitive sequences of monocentrics as those at the centromeres of rye, wheat, *Aegilops* and barley, and also subtelomeric barley repeats may establish, in addition to globular structures also ring-like structures (Schubert et al. 2016). The similar behaviour of *Rg*SAT1/1 and *Rg*SAT1/2 of *R. globosa* confirms that such a ring formation is rather a general feature of repetitive DNA than required for centromere function.

The preferential location of repetitive sequences or heterochromatin in subterminal regions was described as a common feature for holocentric plants (e.g. Guerra and Garcia 2004; Heckmann et al. 2013). It seems to be required for the end-to-end association between homologous non-sister chromatids after metaphase I of *Luzula elegans*, when sister chromatids and not homologues separate (Heckmann et al. 2014). *Rhynchospora* is also undergoing such a type of meiosis, and end-to-end

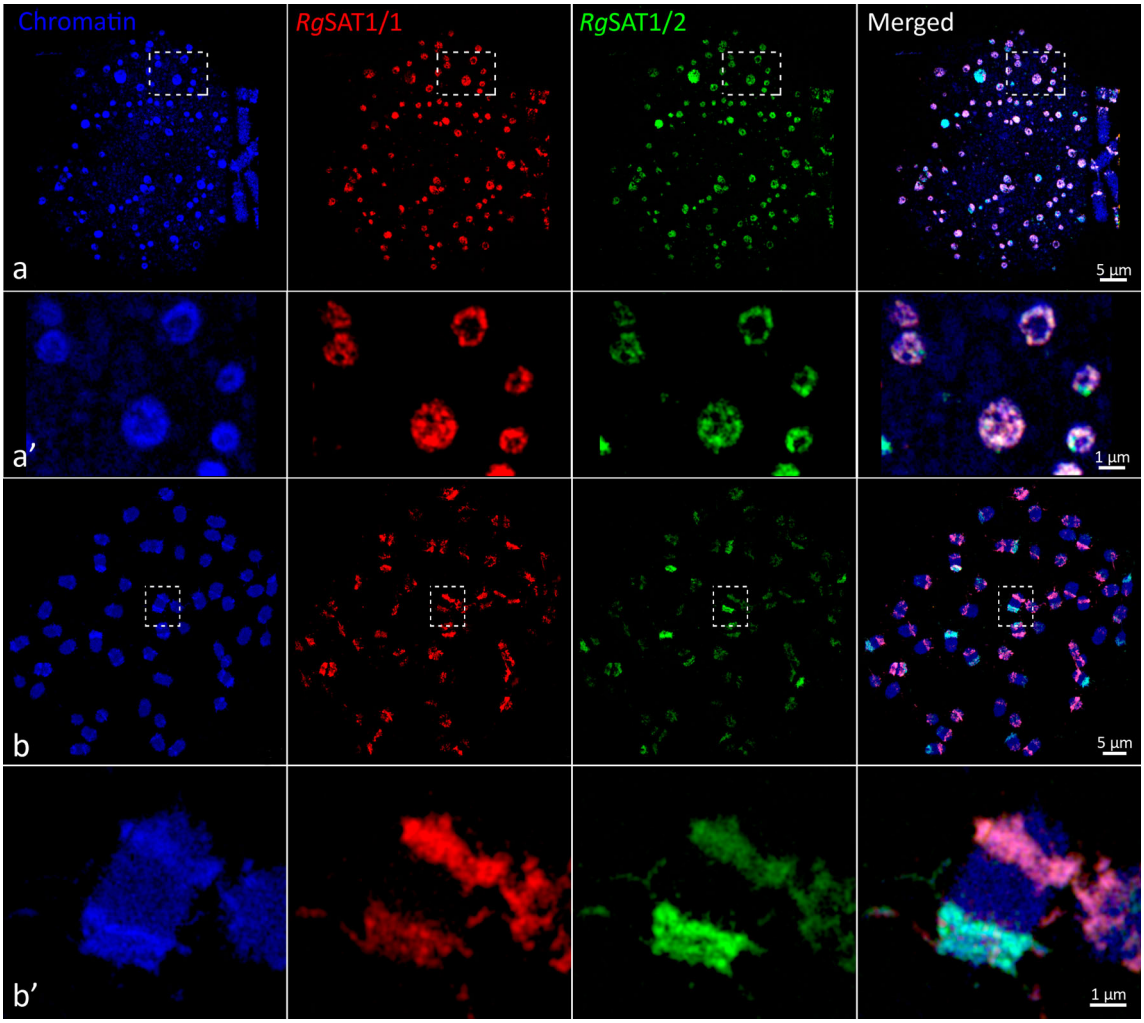


Fig. 5 Distribution of the two most abundant satellite subrepeats of *R. globosa* *Rg*SAT1 analysed by SIM. *Rg*SAT1/1 (*red*) and *Rg*SAT1/2 (*green*) superimposed to chromatin (*blue*) in interphase nuclei (*a*) and metaphase chromosomes (*b*). Both subrepeats localise preferentially to chromocentres (*a*) and chromosome (sub) termini (*b*). In interphase nuclei, both form globular or ring-like structures (*a'*). *Dotted rectangles* indicate enlarged regions in (*a'* and *b'*)

connections of homologues probably exist (Cabral et al. 2014). However, the distribution of repeats along the chromosome showed no preferential sites of clustering, which were visualised both at interstitial and terminal positions in *R. ciliata* and *R. tenuis*. Hence, in this case, satellite DNA might not be involved in the meiotic end-to-end connection of non-sister chromatids. Both *R. globosa* repeats (*Rg*SAT1/1 and *Rg*SAT1/2) were mapped preferentially at the (sub) termini of the smaller and more numerous chromosomes, but no Tyba-like centromeric repeats were found. Hence, a biased distribution of satDNAs at chromosome termini is not a compulsory feature for holocentric species. Nevertheless, it is not clear if the presence of Tyba as a dispersed repeat in *R. tenuis* and *R. ciliata* might be associated with the interstitial distribution of other satDNAs in these species. Investigations of other *Rhynchospora* species will be required to test this possible correlation. The colocalisation of centromeric and non-centromeric repeats resulted in a ladder-like hybridisation pattern along mitotic holocentric chromosomes of *R. ciliata* and *R. tenuis*.

Another cause of a biased terminal localisation of tandem repeats was postulated to explain the terminal 45S rDNA distribution in many holocentric species. It could be explained by the putative deleterious effect of an interstitial, decondensed NOR, which would split the holocentromere into two domains and enable the association of the same chromatid to different poles, as a 'di-holocentric' chromosome (Heckmann et al. 2013). However, this scenario would not apply to other tandemly organised sequences, such as most of the satDNAs reported here, that remain condensed during whole mitosis.

The analysis of the satDNA repeats found in *Rhynchospora* also revealed a further type of chromatin organisation. In addition to the satellite families nested within heterochromatic regions, *Rc*SAT20 and *Rt*SAT239 formed signal clusters at interphase and metaphase but were not associated with

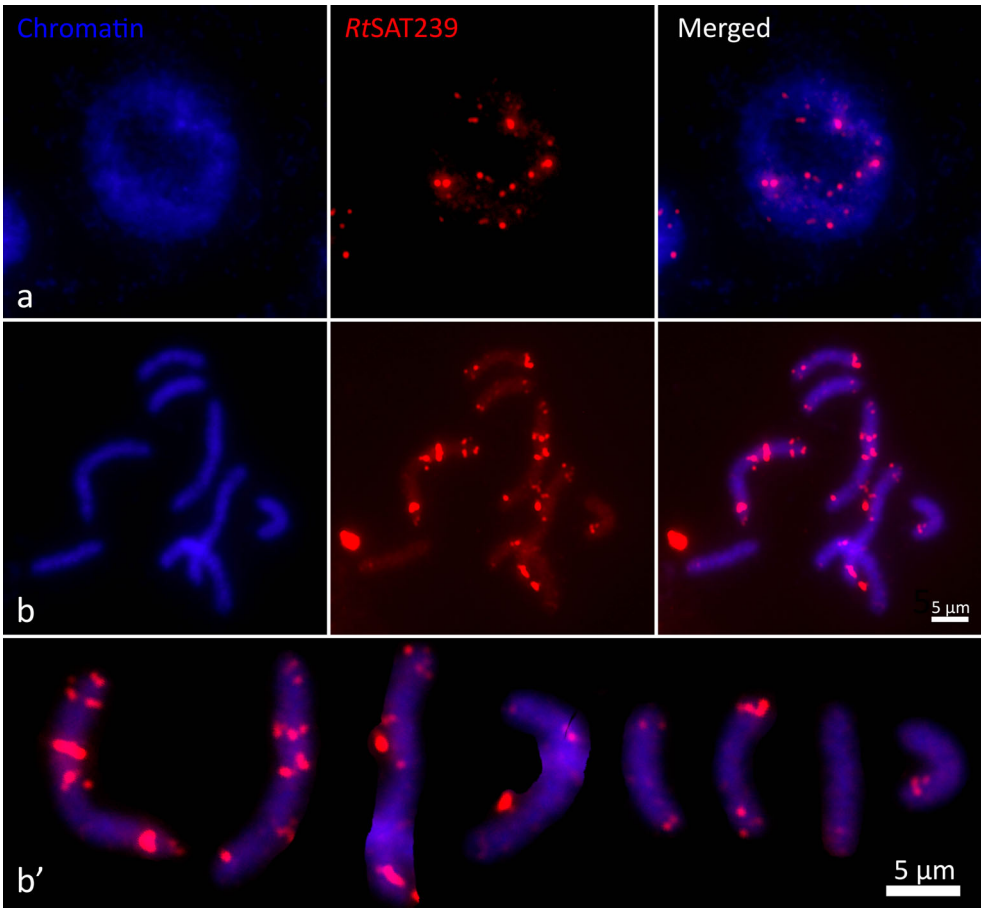


Fig. 6 Interphase nucleus (*a*) and mitotic metaphase chromosomes (*b*) of *R. tenuis* ($2n = 4x = 8$) showing *Rt*SAT239 clusters. Individual chromosomes from (*b*), showing unequal hybridisation patterns among homologues (*b'*)

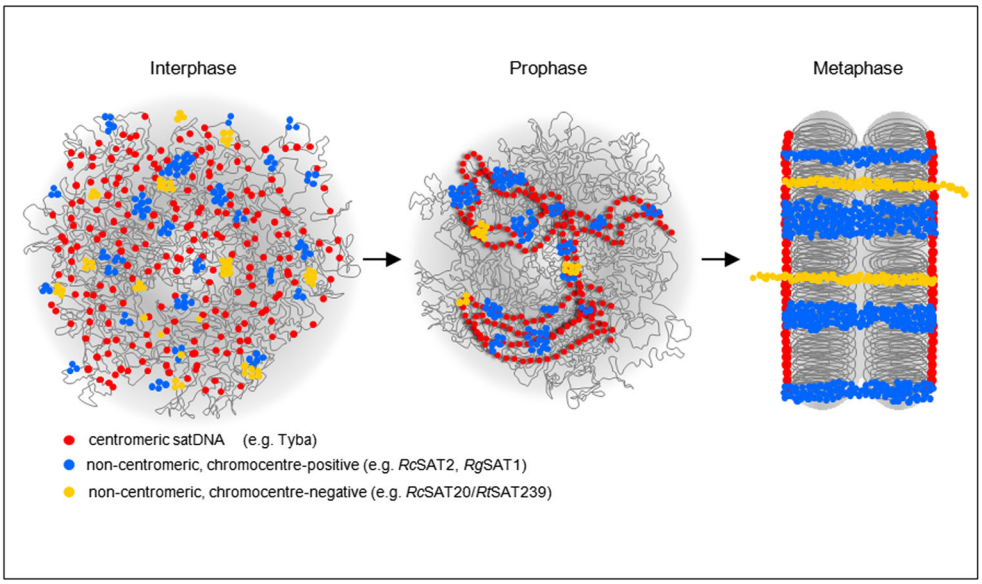


Fig. 7 SatDNAs dynamics during the somatic cell cycle of holocentric *Rhynchospora* species. During interphase, centromeric satDNA arrays (*red*) dissociate into individual CENH3/centromere repeat-containing units. Non-centromeric satDNAs form chromocentre-positive (*blue*) or chromocentre-negative (*yellow*) clusters. Then, in pro- and metaphase, the centromeric satDNAs reassociate and form the holocentromeres along the sister chromatids. In contrast, non-centromeric satDNAs form bands, perpendicularly to the holocentromeres. Non-centromeric chromocentre-negative satDNA may form decondensed chromatin loops at metaphase chromosomes. The model is based on the data derived from *R. ciliata*, *R. globosa* and *R. tenuis*

chromocentres. This is particularly evident in the interphase of *R. ciliata*. Furthermore, at somatic metaphase, they were also present within chromatin loops outside of the condensed chromosomes. Previously, such a decondensed chromatin has been observed only for the active 45S rDNA sites in *Rhynchospora* (Sousa et al. 2011), due to the transcription of ribosomal cistrons (Huang et al. 2008). It is possible that these repeats possess distinct epigenetic modifications, evidencing the presence of different ways to condense satellite DNA during mitosis (Fig. 7). It is not yet known whether the chromatin outlooping is related to transcription as known from animal lampbrush chromosomes (Morgan 2002; Macgregor 2012).

In general, the DNA composition is highly similar between homologous chromosomes. Consequently, after FISH using high- or low-copy probes, the hybridisation patterns of both homologues are usually identical, although exceptions are known for different tandem repeats in monocentric species (e.g. Tek et al. 2005; Pedrosa-Harand et al. 2006; Ferreira et al. 2012). Here, based on the metaphase patterns of non-centromeric satDNA, together with 45S rDNA, we could not identify the chromosome pairs of *R. ciliata* due to the lack of matching labelling patterns. A similar situation was also observed for the distribution of *Rt*SAT239 in the putative autotetraploid *R. tenuis*. It is possible that the differences in number and size of the clusters between homologues are caused by a high amount of heterozygosity, due to the lack or low frequency of meiotic recombination in these species. This has been shown for *R. tenuis* (Cabral et al. 2014), for *R. pubera* (Cabral et al. 2014) and is possibly also true for *R. ciliata*. Low recombination could lead to a differential accumulation of sequences at homologues, as e.g. observed for sex chromosomes (reviewed by Hobza et al. 2015). Alternatively, but less likely, beside complex chromosome rearrangements, a differential degree of condensation in the homologue during mitosis could result in an apparently different distribution of satDNAs between homologues.

In summary, the comparative analysis of satellite repeats in *Rhynchospora* revealed a highly dynamic scenario. While centromeric Tyba satDNA appeared as a more conserved sequence present in two of the investigated species along the holocentromeres, the non-centromeric satDNAs in *R. ciliata*, *R. globosa* and *R. tenuis* are more complex. Repeats differing in their DNA composition and distribution within and among species were found. While centromeric repeats compose linear structures along the mitotic chromatids and disperse at interphase, non-centromeric repeats form distinct clusters along mitotic chromosomes, which may be associated with chromocentres or not.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no competing interests.

Research involving human participants and/or animals No research involving human participants or animals was performed.

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